

Decision Making for Autonomous Driving Via Multimodal Transformer and Deep Reinforcement Learning*

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Abstract— On the basis of environmental information processed by the sensing module, the decision module in automatic driving integrates environmental and vehicle information to make the autonomous vehicle produce safe and reasonable driving behavior. Considering the complexity and variability of the driving environment of autonomous vehicles, researchers have begun to apply deep reinforcement learning (DRL) in the study of autonomous driving control strategies in recent years. In this paper, we apply an algorithm framework combining multimodal transformer and DRL to solve the autonomous driving decision problem in complex scenarios. We use ResNet and transformer to extract the features of LiDAR point cloud and image. We use Deep Deterministic Policy Gradient (DDPG) algorithm to complete the subsequent autonomous driving decision-making task. And we use information bottleneck to improve the sampling efficiency of RL. We use CARLA simulator to evaluate our approach. The results show that our approach allows the agent to learn better driving strategies.

I. INTRODUCTION

The field of autonomous driving has received extensive attention from researchers. However, the complex and uncontrollable driving environment has brought great challenges to autonomous driving decision-making.

Autonomous driving decision-making is a difficult task because of its high requirements for safety on the one hand and uneven distribution among different driving scenarios on the other hand. At present, there are two main categories in the field of autonomous driving decision: rule-based method and learning-based method. Rule-based methods usually require a large number of manually designed rules and a large amount of expert knowledge. Facing complex driving scenarios that have never appeared before, how to make correct and reliable decision-making behaviors is a challenge for traditional rule-based methods [1], [2]. The method based on supervised learning only considers the normal driving situation [3]. Therefore, neither of these two methods can handle the autonomous driving decision-making tasks very well.

In the past few years, deep reinforcement learning (DRL) has made breakthroughs in the fields of games [4], robot navigation [5], and autonomous driving [6]. The advantage of DRL is that it combines the powerful understanding ability of deep learning in perception based on raw data, and the

decision-making ability of RL based on low-dimensional features. The method based on DRL automatically learns strategies through interaction with the environment and is suitable for driving decision-making tasks.

RGB camera is one of the main sensors for sensing driving environment [7], [8], although cameras provide dense perceptual information about the scene, but they lack reliable 3D information and are vulnerable to the weather change. LiDAR can capture 3D information about the environment, but LiDAR measurements are often sparse (especially at a distance) and fail to capture important information such as the status of traffic lights. Therefore, camera-only and LiDAR-only methods are not applicable in some complex urban environments.

In order to solve the above problems, we adopt an algorithm framework based on the combination of multimodal transformer and DRL as shown in Figure 1 in this paper. We introduce ResNet and Transformer to extract the global features of LiDAR point clouds and images. And we use the information bottleneck to improve the sampling efficiency of RL. The main contributions of this paper are as follows: (1) We use a framework that combines multimodal transformer and DRL. And the fusion attention features obtained by ResNet and multimodal transformer is visualized to increase the interpretability of the framework. (2) We use DDPG algorithm with information bottleneck in simulation environment CARLA to complete decision-making tasks. The information bottleneck is used to improve the sampling efficiency of RL. And the results show that our method is better.

II. RELATED WORKS

A. Camera-based Autonomous Driving by Supervised Learning

Bojarski *et al.* [3] used human driving data to train an end-to-end convolutional neural network that maps raw pixel images captured by the vehicle's front cameras to steering commands. But they need to collect training data by driving on a wide variety of roads and in a variety of weather conditions, and they can only do simple tasks by controlling the steering angle. A recent approach to solving autonomous driving tasks is imitation learning (IL) [9], [10], which aims to learn control policies by observing demonstrations by experts. But imitation learning also has challenges in scaling. It doesn't have access to expert instances to imitate every potential situation one might encounter.

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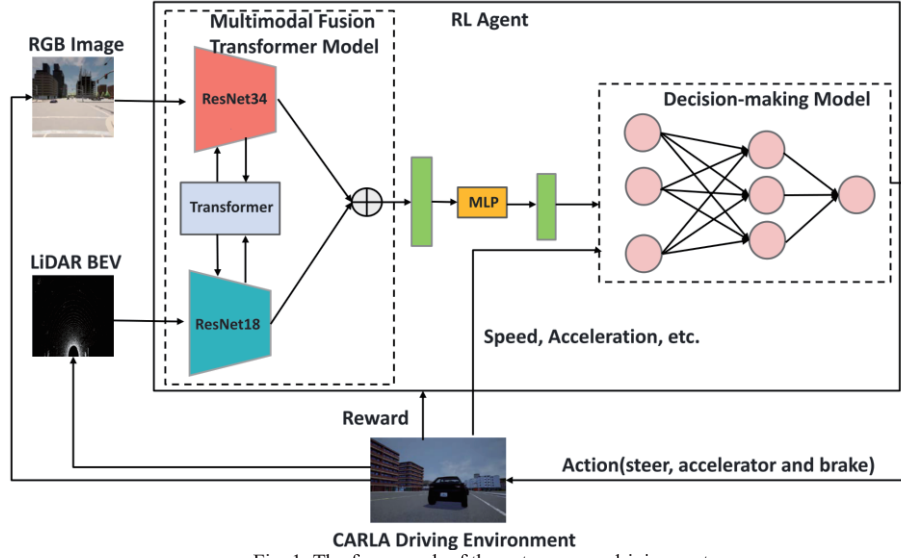


Fig. 1. The framework of the autonomous driving system.

B. Camera-based Autonomous Driving by Reinforcement Learning

Alex Kendall *et al.* [11] demonstrated the first application of DRL in autonomous driving. With randomly initialized parameters, their model uses a monocular image as input and is able to learn a lane tracking policy with only a small number of training scenarios. Chen *et al.* [12] proposed an interpretable deep reinforcement learning method for end-to-end autonomous driving, which is able to handle complex urban scenarios including lane keeping, pedestrian and vehicle avoidance, and traffic light detection. However, the above methods only use images as input. The camera has a limited viewing angle and lacks 3D information, and cannot accurately perceive environmental information in some special cases. Thus, we can solve this problem by fusing the information from camera and LiDAR sensor, because the characteristics of camera and LiDAR are complementary.

C. Transformer for Autonomous Driving

CNN is widely used in various tasks in computer vision, such as classification, detection, and segmentation. CNN extracts the features by sharing convolution kernels, but its receptive field is usually small, which is not conducive to capturing global features. Chen *et al.* [13] used attention augmented convolutional networks to extract RGB image features.

The transformer [14] contains a multi-head self-attention layer and a fully connected layer that can capture the global information of a picture, so it surpasses many CNN structures in many visual tasks. The Vision Transformer (ViT) proposed by Alexey *et al.* [15] is the first alternative to CNN, using a pure transformer structure. ViT divides the input image into several non-overlapping patches, and then feeds these patches into stacked multiple transformer encoders. However, the computation of global attention by the transformer will lead to huge computational loss, so some methods propose to combine transformer and CNN. Prakash *et al.* [16] combined the transformer and CNN, that is, the feature map of the intermediate layer of ResNet is used as the input of the transformer. In addition, transformer is suitable for

multi-modal tasks. Inspired by [16], we use transformer to extract the features of the image and LiDAR point cloud, and then feed it to the reinforcement learning decision-making network.

III. METHODS

A. Multimodal Fusion with Transformer

For image input, we consider the range 100° of the front camera and get the image size of 256×256 pixels. For the lidar point cloud input, to avoid direct processing of LiDAR point cloud with too much computation, we first convert LiDAR point cloud into a BEV. LiDAR BEV is obtained by the projection of lidar point cloud on XY coordinate plane after discretization, in which the resolution of discretization needs to be specified artificially. We convert the LiDAR point cloud into a 2-bin histogram over a 2D BEV grid with a resolution of 256×256 pixels. For the histogram, we divide it into two intervals in the height dimension, namely the points above and below the datum, and finally produce a 256×256 pixel two-channel pseudo-image. As shown in Fig.2, our key idea is to use transformer-based feature fusion to enable the network to aggregate features from the relevant regions of the two sensors (camera and LiDAR) and to capture global information of the entire 3D scene. We take RGB images and LiDAR BEV as inputs to the model, which are then processed by a convolutional neural network (ResNet model) to produce intermediate feature maps of different resolutions. Transformer combines the image and LiDAR features in a variety of resolutions. Finally, a feature vector is output and passed to the subsequent decision network.

We treat the intermediate feature map of each modality as a set, where each element is treated as a token. Let the intermediate feature map of each modality be a tensor of $H \times W \times C$ and for both modalities these features are superimposed together to form a sequence of $(2 * H * W) \times C$. Suppose the size of the feature map is expressed as $X \in \mathbb{R}^{H \times W \times D_f}$, where H , W and D_f represent the height, width and number of channels of the feature map respectively;

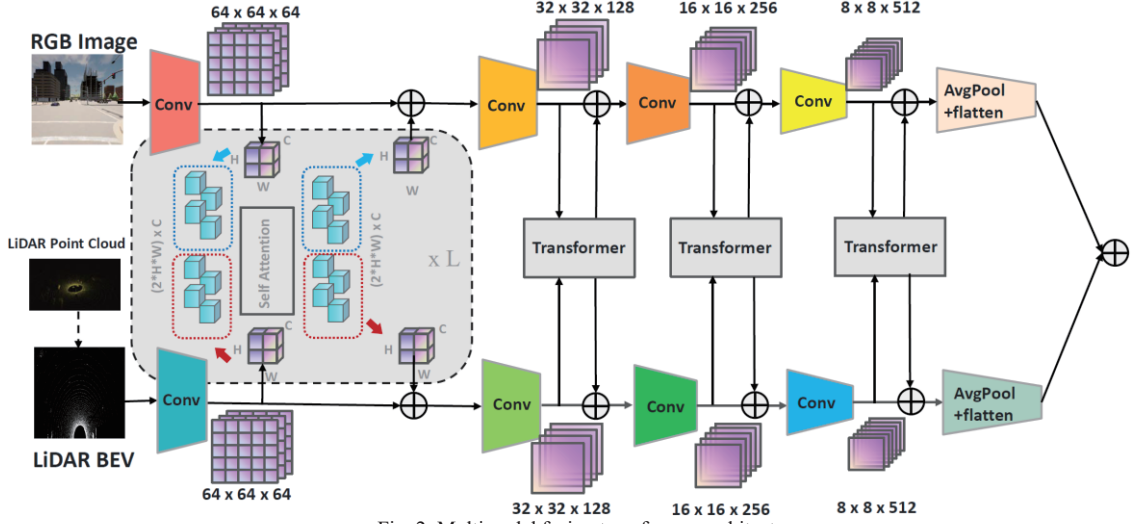


Fig. 2. Multimodal fusion transformer architecture.

then we convert the feature map into a matrix form: $X \in \mathbb{R}^{HW \times D_f}$, where HW represents the number of tokens in the sequence, and each token is composed of a feature vector of dimension D_f . The self-attention mechanism calculates queries, keys and values (Q, K, V) by linear projections:

$$Q = XW^q, K = XW^k, V = XW^v \quad (1)$$

where $W^q \in \mathbb{R}^{D_f \times D_q}, W^k \in \mathbb{R}^{D_f \times D_k}, W^v \in \mathbb{R}^{D_f \times D_v}$ are weight matrices. Then use the scaled dot product to calculate the attention weights between Q and K, and then aggregate the values of each query to get the attention feature $SA \in \mathbb{R}^{HW \times D_v}$:

$$SA = \text{Softmax} \left(\frac{QK^T}{\sqrt{D_k}} \right) V \quad (2)$$

In order to extract the complete attention information, we adopt a multi-head attention mechanism. First divide the (Q, K, V) obtained in Eq. (1) into h subspaces along the dimension of the channel:

$$\begin{aligned} Q &= [Q_1, \dots, Q_i, \dots, Q_h] \\ K &= [K_1, \dots, K_i, \dots, K_h] \\ V &= [V_1, \dots, V_i, \dots, V_h] \end{aligned} \quad (3)$$

So the multi-head self-attention mechanism can be obtained:

$$\begin{aligned} SA_i &= \text{Softmax} \left(\frac{Q_i(K_i)^T}{\sqrt{D_k^i}} \right) V_i \\ MSA &= \text{Concat}(SA_1, \dots, SA_i, \dots, SA_h)W^o \end{aligned} \quad (4)$$

where $W^o \in \mathbb{R}^{D_v \times D_o}$ is a parameter matrix.

When performing attention feature extraction, the relative positional relationship between pixels needs to be considered. The self-attention model realizes attention extraction through matrix dot product, and the relative positional relationship between pixels is not considered in the calculation process. Inspired by [17], we encode the relative position of the feature map in the self-attention model. We compute the information encoding between any two pixels ($i(x_i, y_i), j(x_j, y_j)$) of

feature maps in both width and height directions respectively. Then the relative position encoding matrix of the feature map can be expressed as $D_x \in \mathbb{R}^{HW \times HW}, D_y \in \mathbb{R}^{HW \times HW}$. After adding the relative position information of the pixels, the multi-head self-attention can be expressed as

$$SA_i = \text{Softmax} \left(\frac{Q_i(K_i)^T + D_x^i + D_y^i}{\sqrt{D_k^i}} \right) V_i \quad (5)$$

$$MSA = \text{Concat}(SA_1, \dots, SA_i, \dots, SA_h)W^o$$

At different resolutions, the output features can be expressed as

$$F^{out} = \text{MLP}(MSA) + X \quad (6)$$

B. Decision-making Model based on DRL with Information Bottleneck

We use model-free and off-policy DDPG algorithm with information bottleneck to complete decision-making tasks. First, we model the problem as a Markov Decision Process represented by $\langle s_t, a_t, p, r_t, \gamma \rangle$, where s_t is the states composed of the fusion attention features and other information (speed, acceleration, etc.), a_t is the control actions of vehicle combined with steer, accelerator and brake, p is the state transition probability, r_t is the rewards for the vehicle obtained through the environment and γ is the discount factor. The goal of reinforcement learning is to maximize the discounted cumulative rewards.

In this paper, the critic network is $Q(s, a | \theta^Q)$ with weights θ^Q and the actor network is $\mu(s | \theta^\mu)$ with weights θ^μ . The critic network updates network parameters by minimizing the MSE between the target Q value y_i and the actual value $Q(s_i, a_i | \theta^Q)$:

$$L = \frac{1}{N} \sum_i (y_i - Q(s_i, a_i | \theta^Q))^2 \quad (7)$$

$$y_i = r_i + \gamma Q'(s_{i+1}, \mu'(s_{i+1} | \theta^{\mu'}) | \theta^Q)$$

where $\gamma = 0.95$ and μ' is the target actor network.

Update the actor policy using the sampled policy gradient:

$$\nabla_{\theta^{\mu}} J \approx \frac{1}{N} \sum_i \nabla_a Q(s, a | \theta^Q) \bigg|_{s=s_i, a=\mu(s_i)} \nabla_{\theta^{\mu}} \mu(s | \theta^{\mu}) \bigg|_{s_i} \quad (8)$$

Then update the target networks by soft update method.

To improve the training efficiency of the model, we use the prioritized experience replay to sample the historical data. In addition, as shown in Fig.3, we introduce information bottleneck to optimize reinforcement learning strategies by minimizing the objective function:

$$L_{IB} = H(Y|Z) + \beta I(Z, S) \quad (9)$$

$$I(Z, S) = E_{p(S)} [D_{KL}[p(Z|S) \| q(Z)]]$$

where S is the input state, which refers to the image and LiDAR point cloud, Z is the representation information, which refers to the attention features, Y is the output of the reinforcement learning model, $H(\cdot)$ is the information entropy, $I(\cdot)$ is the mutual information, $D_{KL}(\cdot)$ is the KL divergence and $q(\cdot)$ is the standard normal distribution. Combined with the target function in the DDPG algorithm, we can get

$$H(Y|Z) = L(\theta^Q) + L(\theta^{\mu}) \quad (10)$$

Finally, we get the target function for policy updates to the multimodal fusion transformer model:

$$L_{IB} = H(Y|Z) + \beta I(Z, S) \quad (11)$$

$$= L(\theta^Q) + L(\theta^{\mu}) + \beta E_{p(S)} [D_{KL}[p(Z|S) \| q(Z)]]$$

where β is a Lagrange multiplier.

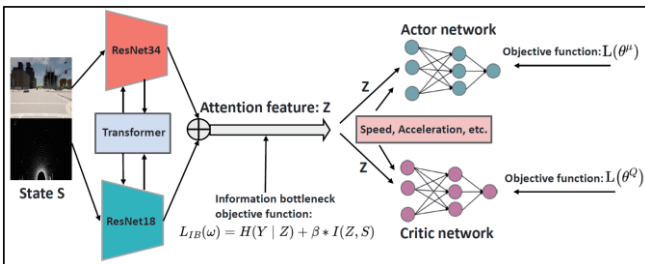


Fig. 3. DDPG algorithm framework based on information bottleneck.

IV. EXPERIMENTS

We use CARLA simulation environment [18] to verify the performance of the proposed model, which is an open-source urban driving simulator with a variety of scenarios as shown in Fig.4. We choose the Town03 to train our model, which is the most complex town, with a 5-lane junction, a roundabout, unevenness, a tunnel, and more.

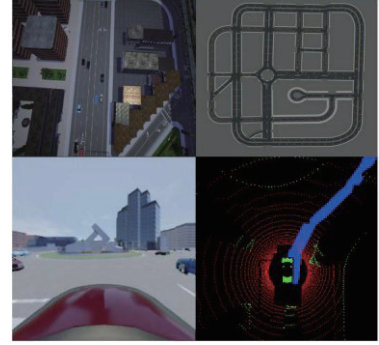


Fig. 4. CARLA driving environment

A. Implementation details

We use two sensor modalities-the RGB image encoded by ResNet-34 and the LiDAR point cloud encoded by ResNet-18. And we use 1 transformer at different resolutions, and each transformer has 4 attention heads. In addition, we design the reward function and specify metrics to compare the performance of different methods.

1) Reward Function

We mainly consider three aspects: safety, driving efficiency and comfort. For safety, we design the following reward function:

$$R_{collision} = \begin{cases} R_{near-collision}, & \text{if } D < 20\text{m} \\ -200, & \text{if collision} \end{cases} \quad (12)$$

$$R_{out} = -abs(d)$$

$$R_{angle} = -abs(\theta)$$

where D is the distance between the vehicle and the obstacle, d is the distance that the vehicle deviates from the current lane, the θ is the angle between the vehicle and the lane. For driving efficiency, the speed of the vehicle is within the allowable range as fast as possible:

$$R_{speed} = \begin{cases} v, & \text{if } v \leq v_d \\ v_d - v, & \text{if } v > v_d \end{cases} \quad (13)$$

where v_d is the maximum speed limit. For comfort, we limit the variable acceleration in both lateral and longitudinal directions:

$$R_{comf} = -\alpha \cdot \dot{a}_x^2 - \beta \cdot \dot{a}_y^2 \quad (14)$$

where $\dot{a}_x, \dot{a}_y$ are the lateral and longitudinal variable accelerations respectively. Finally, the total reward function is

$$R_{total} = w_1 R_{collision} + w_2 R_{out} + w_3 R_{angle} + w_4 R_{speed} + w_5 R_{comf} \quad (15)$$

where w_1, w_2, w_3, w_4, w_5 are weight factors. And we set $w_1, w_2, w_3, w_5 = 1, w_4 = 4$

2) Metrics

We specify three metrics. The first is route completion R_i , which is the percentage of the route distance completed by a vehicle. The second is infraction penalty P_i . Ordered by severity, the infractions are the following: collisions with pedestrians-0.40, collisions with other vehicles-0.55.

collisions with static elements-0.60, running a red light-0.80. The third is driving score $R_i P_i$.

B. Results

We first use the total reward and metrics to compare the effects of DDPG under two frameworks: multimodal fusion with transformer and camera-only with self-attention. As shown in Fig.5 and Table 1, we get the convergent total reward after 600 episodes and our method is better than the other one, which shows that multimodal fusion transformer model can extract more key feature



Fig. 5. The total reward of two models (multimodal fusion and image-only)

TABLE I. DRIVING PERFORMANCE

Methods	Metrics	
	Route Completion	Driving Score
Multimodal fusion	85.80	55
Camera-only	74.20	43

As shown in Fig. 6 and Table2, we trained 3750 episodes to compare four models: multimodal transformer with information bottleneck, multimodal fusion, image-only with self-attention and image-only with CNN. The result shows that our method is better.

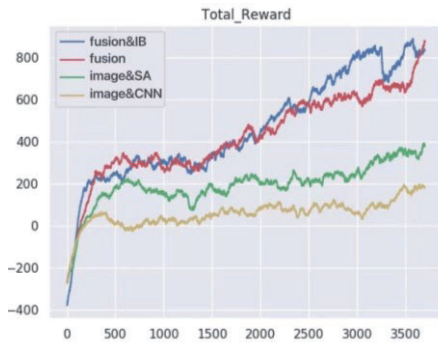


Fig. 6. The total reward of four models (multimodal fusion with information, multimodal fusion, image-only with self-attention and image-only with CNN)

TABLE II. DRIVING PERFORMANCE

Methods	Metrics	
	Route Completion	Driving Score
Multimodal fusion with IB	87.30	57

Methods	Metrics	
	Route Completion	Driving Score
Multimodal fusion	85.30	54
Image-only with SA	73.85	41
Image-only with CNN	69.44	36

By visualizing the fused attention features of the multimodal fusion transformer model, we can clearly see the effect of the model extracting features at different resolutions and improve the interpretability of our method.

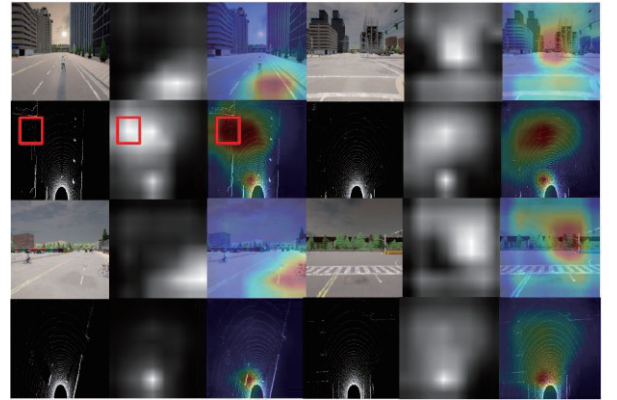


Fig. 7. Attention maps. The first and fourth columns are the raw information obtained by camera and LiDAR. The second and fifth columns are attention maps. The third and sixth columns are the color attention maps.

As shown in Fig.7, it is the visualizations of attention maps obtained in some scenes of Town05. We can clearly see that the vehicle can detect the obstacles (pedestrians, vehicles) and traffic lights in front. Due to the limited viewing range of the camera, we can detect a wider range of obstacle information after adding LiDAR, so as to ensure that vehicles, especially in the intersection scene, can notice vehicles coming from the left and right. For example, in the first scene on the left in the figure, in fact, there is a car (red box) at the left intersection of the vehicle. The camera only notices the pedestrians in front, while the lidar notices the vehicles on the left. By fusing the information of images and lidar feature maps, we can obtain the global information of the whole environment and ensure driving safety.

V.CONCLUSION

In this paper, we use an algorithm framework based on multimodal fusion transformer model and DRL with information bottleneck for autonomous driving decision-making in CARLA simulator. The results show that our method can find better driving strategies with higher efficiency.

In the future, we will consider using safe reinforcement learning to improve the rationality and safety of autonomous driving decisions.

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